N. Platform Operations & Control Plane

Standalone capability brief · Suggested module: **Platform Control Plane**

Purpose

Manage the lifecycle of platform resources independently from the data-plane traffic that uses them, especially for PaaS products that provision and operate customer environments.

1. Scope and capability areas

Provisioning

Create, configure, and retire tenants, projects, databases, services, and related resources through repeatable workflows.

Inventory and discovery

Maintain resource inventory, ownership, dependencies, state, and service registration.

Placement and capacity

Choose regions, zones, hosts, and capacity pools according to policy and availability.

Health and maintenance

Standardize readiness, health checks, maintenance windows, and planned downtime communication.

Upgrade and rollback

Orchestrate staged upgrades, compatibility checks, and rollback or recovery paths.

Backup and disaster recovery

Coordinate backups, restores, failover, and verification against recovery objectives.

Status and incidents

Expose platform status, incident communication, and resource-level operational state.

2. Shared contracts and standards

- Every resource has a desired state, observed state, owner, tenant, region, lifecycle state, and reconciliation policy.
- Control-plane operations are idempotent and recoverable after interruption.
- Data-plane requests should not depend synchronously on a healthy control plane for every ordinary operation.
- Resource placement and regional policies must be explicit and auditable.

3. Recommended implementation approach

- Use declarative desired state and reconciliation loops for long-lived resources.
- Model provisioning, upgrade, and recovery as durable workflows.
- Separate management APIs from customer data traffic and apply different security controls.
- Track dependencies and blast radius before maintenance or rollout.
- Test region loss, partial provisioning, failed upgrades, and control-plane unavailability.

4. Failure behavior and operational expectations

- Partial provisioning must remain visible and be resumable or safely cleaned up.
- Control-plane outage should not unnecessarily interrupt healthy data-plane traffic.
- Rollback limitations must be documented; irreversible operations need explicit safeguards.

5. Suggested operational metrics

- Provisioning lead time and success rate
- Desired/observed state drift
- Upgrade failure and rollback rates
- Recovery objective attainment
- Control-plane availability and queue age

6. Readiness and acceptance checklist

- ■ Resources can be provisioned and recovered through repeatable workflows.
- ■ Desired and observed state drift is detected and reconciled.
- ■ Control-plane outages have bounded impact on data-plane traffic.
- ■ Maintenance and upgrades include safety checks and rollback planning.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.